Labsheet 3 SECS05-AS03

Mastery in Data Analytics and Visualizations and Career Advancement

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In [ ]:
```
# Khushi Choudhary CU240251564
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import statsmodels.api as sm
from statsmodels.formula.api import ols

# Load dataset
df = pd.read_csv('WA_Fn-UseC_-HR-Employee-Attrition.csv')
# Quick inspection
print("Dataset Shape:", df.shape)
print(df.head())
```

```
Dataset Shape: (1470, 35)
   Age Attrition     BusinessTravel  DailyRate              Department  \
0   41       Yes      Travel_Rarely       1102                   Sales
1   49        No  Travel_Frequently        279  Research & Development
2   37       Yes      Travel_Rarely       1373  Research & Development
3   33        No  Travel_Frequently       1392  Research & Development
4   27        No      Travel_Rarely        591  Research & Development

   DistanceFromHome  Education EducationField  EmployeeCount  EmployeeNumber  \
0                 1          2  Life Sciences              1               1
1                 8          1  Life Sciences              1               2
2                 2          2          Other              1               4
3                 3          4  Life Sciences              1               5
4                 2          1        Medical              1               7

   ...  RelationshipSatisfaction StandardHours  StockOptionLevel  \
0  ...                         1            80                 0
1  ...                         4            80                 1
2  ...                         2            80                 0
3  ...                         3            80                 0
4  ...                         4            80                 1

   TotalWorkingYears  TrainingTimesLastYear WorkLifeBalance  YearsAtCompany  \
0                  8                      0               1               6
1                 10                      3               3              10
2                  7                      3               3               0
3                  8                      3               3               8
4                  6                      3               3               2

  YearsInCurrentRole  YearsSinceLastPromotion  YearsWithCurrManager
0                  4                       0                     5
1                  7                       1                     7
2                  0                       0                     0
3                  7                       3                     0
4                  2                       2                     2

[5 rows x 35 columns]
```

In [ ]:
```
numerical_cols = ['Age', 'MonthlyIncome', 'TotalWorkingYears', 'YearsAtCompany']

# Descriptive summary table
```

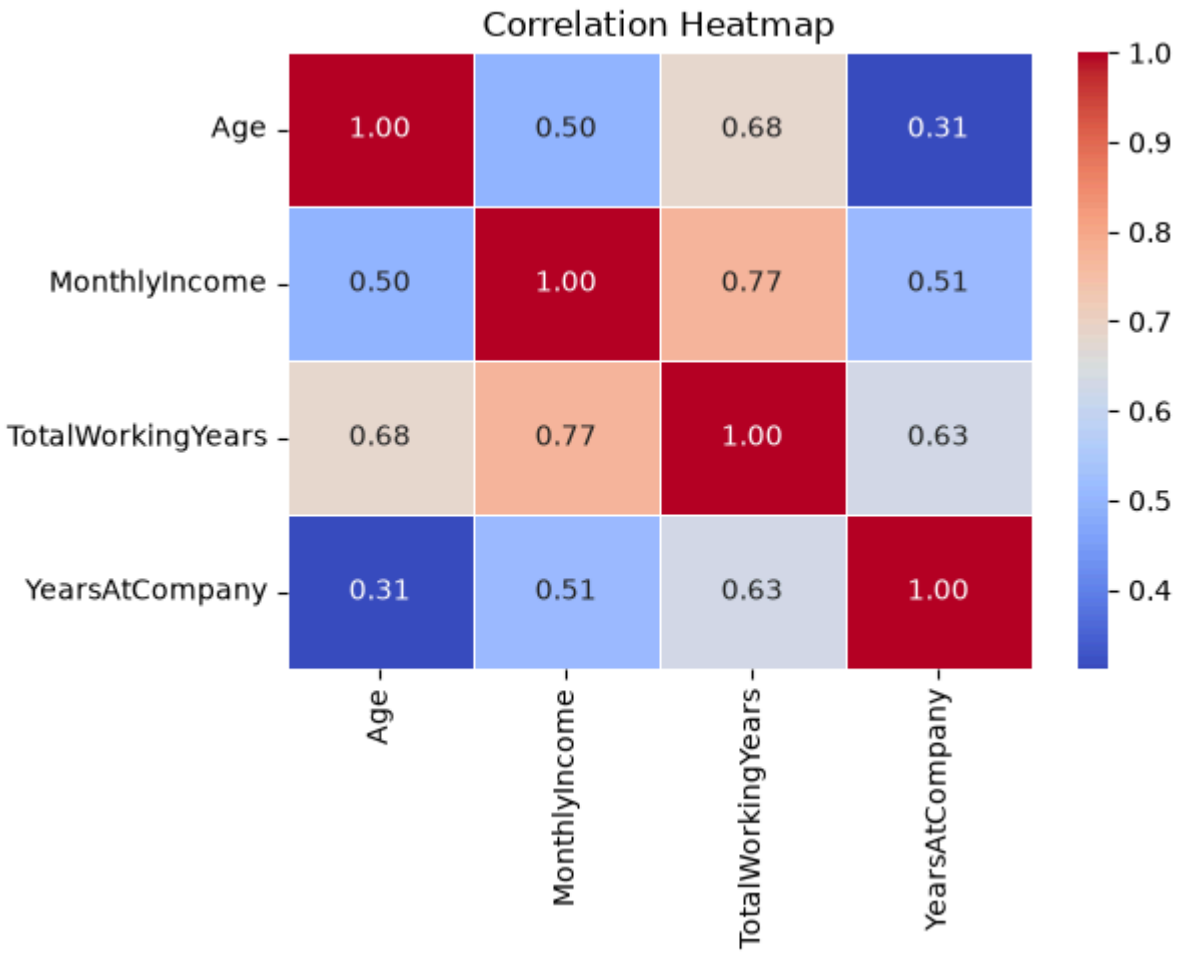
```
desc_stats = pd.DataFrame({
    'Mean': df[numerical_cols].mean(),
    'Median': df[numerical_cols].median(),
    'Mode': df[numerical_cols].mode().iloc[0],
    'Variance': df[numerical_cols].var(),
    'Std Dev': df[numerical_cols].std()
})
print("--- Descriptive Statistics ---")
print(desc_stats)
```

```
--- Descriptive Statistics ---
                          Mean  Median  Mode      Variance      Std Dev
Age                  36.923810    36.0    35  8.345505e+01     9.135373
MonthlyIncome      6502.931293  4919.0  2342  2.216486e+07  4707.956783
TotalWorkingYears    11.279592    10.0    10  6.054056e+01     7.780782
YearsAtCompany        7.008163     5.0     5  3.753431e+01     6.126525
```

In [8]:
```
# Pearson correlation matrix
corr_matrix = df[numerical_cols].corr(method='pearson')
print("--- Pearson Correlation Matrix ---")
print(corr_matrix)

# Heatmap visualization
plt.figure(figsize=(6, 4))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title('Correlation Heatmap')
plt.show()
```

```
--- Pearson Correlation Matrix ---
                          Age  MonthlyIncome  TotalWorkingYears  YearsAtCompany
Age                  1.000000       0.497855           0.680381        0.311309
MonthlyIncome        0.497855       1.000000           0.772893        0.514285
TotalWorkingYears    0.680381       0.772893           1.000000        0.628133
YearsAtCompany       0.311309       0.514285           0.628133        1.000000
```



In [ ]:
```
group_yes = df[df['Attrition'] == 'Yes']['MonthlyIncome']
group_no = df[df['Attrition'] == 'No']['MonthlyIncome']
# Two-sample t-test (Welch's t-test for unequal variances)
t_stat, p_val = stats.ttest_ind(group_yes, group_no, equal_var=False)

print("--- Independent Samples t-Test ---")
print(f"t-statistic: {t_stat:.4f}")
print(f"p-value: {p_val:.4e}")

alpha = 0.05
if p_val < alpha:
    print("Decision: Reject H0 -> There is a statistically significant difference in Monthly I
else:
    print("Decision: Fail to Reject H0 -> No statistically significant difference found.")
```

```
--- Independent Samples t-Test ---
t-statistic: -7.4826
p-value: 4.4336e-13
Decision: Reject H0 -> There is a statistically significant difference in Monthly Income based
on Attrition.
```

In [6]:
```
# Fit the ANOVA model
model = ols('MonthlyIncome ~ C(JobRole)', data=df).fit()
anova_table = sm.stats.anova_lm(model, typ=2)

print("--- One-Way ANOVA Table ---")
print(anova_table)

p_val_anova = anova_table['PR(>F)'].iloc[0]
if p_val_anova < 0.05:
    print("\nDecision: Reject H0 -> Significant difference in Monthly Income across Job Roles
else:
    print("\nDecision: Fail to Reject H0 -> No significant difference.")
```

```
--- One-Way ANOVA Table ---
                  sum_sq      df           F  PR(>F)
C(JobRole)  2.657098e+10     8.0  810.214054     0.0
Residual    5.989190e+09  1461.0         NaN     NaN

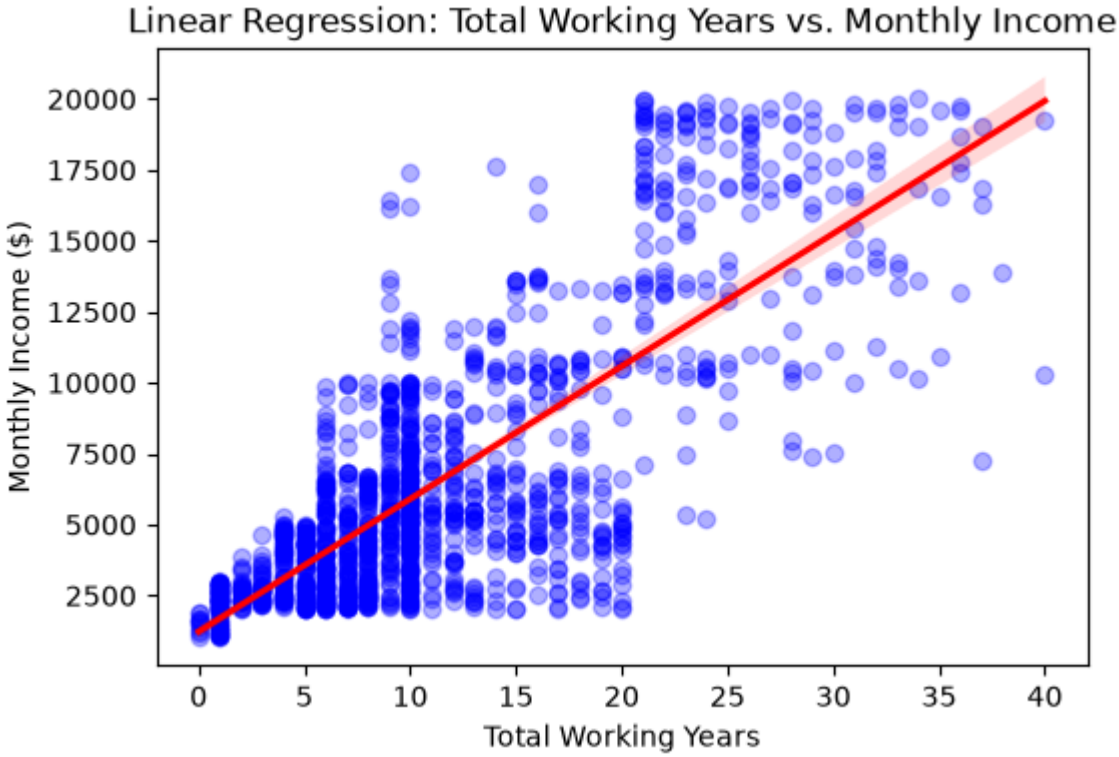
Decision: Reject H0 -> Significant difference in Monthly Income across Job Roles.
```

In [ ]:
```
# Khushi Choudhary CU240251564
# Prepare features
X = df['TotalWorkingYears']
y = df['MonthlyIncome']
# Add constant term (intercept)
X_with_const = sm.add_constant(X)
# Fit OLS Regression
lr_model = sm.OLS(y, X_with_const).fit()
# Print detailed summary
print(lr_model.summary())

# Plot regression line
plt.figure(figsize=(6, 4))
sns.regplot(x='TotalWorkingYears', y='MonthlyIncome', data=df,
            scatter_kws={'alpha':0.3, 'color':'blue'}, line_kws={'color':'red'})
plt.title('Linear Regression: Total Working Years vs. Monthly Income')
plt.xlabel('Total Working Years')
plt.ylabel('Monthly Income ($)')
plt.show()
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          MonthlyIncome   R-squared:                       0.597
Model:                            OLS   Adj. R-squared:                  0.597
Method:                 Least Squares   F-statistic:                     2178.
Date:                Mon, 24 Aug 2026   Prob (F-statistic):          2.73e-292
Time:                        08:23:26   Log-Likelihood:                -13848.
No. Observations:                1470   AIC:                         2.770e+04
Df Residuals:                    1468   BIC:                         2.771e+04
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const              1227.9353    137.299      8.944      0.000     958.612    1497.259
TotalWorkingYears   467.6584     10.021     46.669      0.000     448.002     487.315
==============================================================================
Omnibus:                       47.473   Durbin-Watson:                   1.993
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               79.304
Skew:                           0.269   Prob(JB):                     6.02e-18
Kurtosis:                       4.003   Cond. No.                         24.2
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```



Step 10: Statistical Findings and Business Decision-Making

1. Statistical Summary & Interpretations

Analysis	Metric / Test Statistic	p-value / Value	Statistical Conclusion
Descriptive Stats	$\text{Mean} > \text{Median}$ (Monthly Income)	—	Positively right-skewed; a small number of executives pull the average up.
Correlation	Pearson's r (Years vs. Income)	$r \approx 0.77$	Strong positive linear correlation between total experience and earnings.

Analysis	Metric / Test Statistic	p-value / Value	Statistical Conclusion
Two-Sample t-Test	Welch's t-statistic	$p < 0.05$	**Reject H_0:** Employees who leave earn significantly less than those who stay.
One-Way ANOVA	F-statistic	$p < 0.001$	**Reject H_0:** Mean compensation varies significantly across job roles.
Linear Regression	Slope (β_1) & R^2	$R^2 \approx 0.60$	Total working years explains $\approx 60\%$ of the variance in monthly income.

2. Strategic Business Insights

- **Compensation as an Attrition Driver:**
  Because the difference in mean monthly income between retaining and leaving cohorts is statistically significant ($p < 0.05$), uncompetitive compensation represents a primary risk factor for employee turnover.
- **Role-Based Pay Equity:**
  The ANOVA results confirm wide structural pay tiers across roles. HR should run post-hoc comparisons (e.g., Tukey's HSD) to pinpoint specific underpaid job functions driving attrition.
- **Experience-Linked Compensation Scaling:**
  The regression slope (β_1) provides a baseline benchmark for regular career progression and market-aligned salary adjustments per additional year of experience.

3. Recommendations for Data-Driven Policy

1. **Targeted Retention Reviews:** Conduct compensation reviews specifically for flight-risk roles showing high attrition and below-average pay bands.
2. **Predictive Churn Modeling:** Integrate statistical drivers ($p < 0.05$) into supervised classification models (e.g., Logistic Regression, Random Forest) to predict individual attrition risk early.
3. **Structured Progression Framework:** Use the linear regression formula to formalize predictable, tenure-based compensation bands.